UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/548,314	12/10/2021	Ryan L. Vander Veen	041950/09007	5144

27530 7590 12/23/2024

Nelson Mullins Riley & Scarborough LLP
IP Department
301 S. College Center, Suite 2300
301 S. College Street
Charlotte, NC 28202

EXAMINER	
KIRWIN, STEFANIE JOHANNA	

ART UNIT	PAPER NUMBER
1677	

NOTIFICATION DATE	DELIVERY MODE
12/23/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

ipdocket@nelsonmullins.com

Office Action Summary	Application No. 17/548,314	Applicant(s) Vander Veen et al.	
	Examiner STEFANIE J KIRWIN	Art Unit 1677	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 06/20/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-9 and 20-22 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-9 and 20-22 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 04/25/2022 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Continued Examination Under 37 CFR 1.114

2. A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 06/20/2024 has been entered.

Priority

3. The present application was filed 06/20/2024. The priority information to provisional application No. 63/124,551, filed 12/11/2020 is acknowledged.

Status of the claims

4. Claims 1-9 and 20-22 are pending. Claim 1 is amended, claims 20-22 are new, and claims 10-19 are cancelled. Claims 1-9 and 20-22 are examined below.

New Rejections

Applicants' amendments have necessitated the following ground(s) of rejection:

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

5. **Claims 1-9** and **20-22** are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

Claims 1 and **22** are indefinite because claim 1 recites "the device comprising an aqueous extraction buffer" and claim 22 comprises a recitation of "the aqueous extraction buffer". It is unclear if the device comprises the extraction buffer or whether the liquid sample, which is applied to the device contains the aqueous buffer, in which case the buffer is not part of the device, but rather added to it during a method of detecting an analyte. The specification of the present application refers to the extraction buffer, which can be an aqueous extraction buffer, in regards to the liquid sample preparation (see paragraphs [0047-48], as well as [0014] and [0080]), but not in the context of being part of the device. Therefore, it is unclear which part of the device comprises the aqueous extraction buffer.

Claims 2-9 and 20-22 depend on claim 1 and are therefore also rejected.

Herein, for the purpose of continued examination on the merits, the term "aqueous extraction buffer" is broadly and reasonably interpreted to mean a buffer applied to the device.

Claim 22 contains the trademark/trade name Tween. Where a trademark or trade name is used in a claim as a limitation to identify or describe a particular material or product, the claim does not comply with the requirements of 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph. See *Ex parte Simpson*, 218 USPQ 1020 (Bd. App. 1982). The claim scope is uncertain since the trademark or trade name cannot be used properly to identify any particular material or product. A trademark or trade name is used to identify a source of goods, and not the goods themselves. Thus, a trademark or trade name does not identify or describe the goods associated with the trademark or trade name. In the present case, the trademark/trade name is used to identify/describe Polyethylene glycol sorbitan monolaurate and, accordingly, the identification/description is indefinite.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

6. **Claims 1-4, 6-9, and 20-21 are rejected under 35 U.S.C. 103 as being unpatentable over Caussette et al. (US9182397B2), in view of Watanabe (1998, Analyst), Berlina et al. (2019, Biomedical and Pharmacology Journal), and Godoy-Navajas et al. (2012, Talanta), and Janeway (2001, Immunobiology).**

Regarding **claim 1, 20,** and **21**, Caussette teaches a method of detecting an antibiotic in a sample comprising providing a test device. Caussette further teaches that the test device comprises a solid support comprising a sample receiving region (sample pad), a detection region and an absorbing region (Caussette, column 3, lines 3-7). Caussette further teaches that a labeled antibiotic binding protein and a labeled control reagent are part of the test device and can be present in liquid or solid form and can be located between the sample receiving region and the detection region (conjugate pad). Caussette further teaches a detection zone comprising an immobilized antibiotic test line capable of binding the labeled antibiotic binding protein and a control zone comprising an immobilized binding agent capable of binding the labeled control reagent (control line; Caussette, column 2, lines 23-24). Caussette further teaches labeling the antibiotic binding protein and the control reagent (Caussette, column11, lines 30-31). Caussette further teaches that the binding proteins can be an antibody (Caussette, column 5, lines

1-4). Caussette further teaches that anti-IgY antibody was applied to the control zone (Caussette, column 11, lines 7-9) and that the labeled control reagent is IgY-coated gold particles (Caussette, column 11, lines 46-47). Caussette further teaches a test device having a proximal and a distal end, said test device configured to allow lateral flow from the proximal to the distal end comprising in sequence from the proximal to the distal end a sample receiving region, a detection region and an absorbing region (Caussette, column 2, lines 29-47). Caussette further teaches that the detection zone and control zone were applied onto a nitrocellulose membrane (a membrane surface comprising a test line and a control line) and that the membrane was blocked by spraying a solution comprising Tris buffer (device comprises an aqueous buffer; Caussette, column 11, lines 14).

Caussette fails to teach that the antibiotic is monensin.

Watanabe teaches a monoclonal-based an immunochromatographic rapid assay for monensin (Watanabe, see title). Watanabe further teaches that in the US, there is a cut-off of 0.05 ppm established for monensin in edible tissues in cattle and goats and in Japan 0.05 ppm monensin is considered negligible. Watanabe further teaches that in order to monitor monensin residue levels in livestock products, simple and reliable methods are required (Watanabe, page 2573, lines 7-23).

It would have been *prima facie* obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to have applied the device of Watanabe to test for monensin as taught by Watanabe, because of the teaching of Watanabe that there is a need for a simple and reliable testing method, because there are limits to acceptable levels in edible tissues.

The ordinary artisan would have a reasonable expectation of success, because both Caussette and Watanabe teach success testing monensin in animal tissue using an immunochromatographic assay.

Although Caussette teaches labeled antibiotic binding protein and a labeled control reagent, it does not teach that the conjugate pad is loaded with 30-40 µg/ml anti-monensin specific antibody.

Berlina teaches a lateral flow immunoassay for the detection of antibiotics (Berlina, page 17, Abstract, line 3). Berlina further teaches that in a competitive lateral flow assay the concentration of the specific antibody is one of the most important parameters and that increasing antibody concentration over a concentration that gives an adequate analytical signal increases unspecific staining (Berlina, page 20, 2nd column, see entire 2nd paragraph).

Godoy-Navajas teaches a competitive heterogeneous immunoassay with antibody capture involving immobilization of BSA-monensin. Godoy Navajas further teaches adding 75 µl of 10 nM NP-labeled anti-monensin antibody (Godoy-Navajas, page 197, see '2.3.4. Determination of monensin'). Considering that an IgG antibody is defined by the state of the art as having a molecular weight of approximately 150 kDa (see e.g., Janeway, see 3-1., lines 1-2), Godoy-Navajas teaches adding 20 µg/ml labeled antibody. Regarding the limitations 30-40 µg/ml (claim 1) and 30 µg/ml (claim 20) and 40 µg/ml (claim 21) as presently recited in the claims; it has been previously established by the courts that a *prima facie* case of obviousness exists where the claimed ranges or amounts do not overlap with the prior art but a merely close (see MPEP 2144.05). In the present case, 30-40 µg/ml is an amount substantially close to 20 µg/ml as taught by Godoy-Navajas and as such it would have been considered *prima facie* obvious in view of Godoy-Navajas to have arrived at the presently claimed antibody concentration out of the course of routine experimentation, in order to achieve a desired effect of detecting monensin in a sample with an adequate signal, but without an increase in non-specific binding (see Berlina above).

The courts have also held that a particular parameter, when recognized as a result effective variable, i.e., a variable that achieves a recognized result, may be subject to routine optimization in order to determine optimum or workable ranges of said variable (also MPEP 2144.05). In the present case, it would have been *prima facie* obvious to one of ordinary skill in the art to determine the optimum antibody concentration by optimizing what was known in the art to be a suitable

concentration of antibodies as taught by Godoy-Navajas. Godoy-Navajas teaches that it was routine in the art to use an antibody concentration of 20 µg/ml. It would have been obvious to have arrived at the claimed range between 30 and 40 µg/ml as an obvious matter of routine experimentation because the antibody concentration is an art recognized result effective variable (namely the variable which results in successful detection of monensin without an increase in non-specific signal). It would have further been obvious because of the teaching of Berlina that the concentration of the specific antibody is one of the most important parameters in a competitive lateral flow assay and that increasing antibody concentration over a concentration that gives an adequate analytical signal increases unspecific staining.

One of ordinary skill in the art would have a reasonable expectation of success because an antibody concentration of 20 µg/ml is substantially similar to 30-40 µg/ml. One of ordinary skill would expect success optimizing by routine experimentation because such optimization would allow the operator to achieve the desired detection of monensin and further because optimization is well within the skill level of the ordinary artisan.

Regarding the limitations that the device is a competitive lateral flow assay strip wherein binding of the antibody to the test line results in a detectable signal and the binding of the C antibody to the control line results in a detectable signal and wherein presence of antibiotics in the liquid sample decreases the detectable signal at the test line relative to a sample lacking antibiotics and wherein the LFA strip device has a sensitivity of 10 ppm or less for detecting antibiotic in the liquid sample, these limitations read as method steps performed during the use of the claimed device, such limitations describing what the device does, rather than what it is. As indicated in the citation of Caussette and the combined cited art above, the claimed invention is structurally indistinct from the cited prior art, and as such, would be expected capable of the limitations, as recited.

Regarding **claims 2** and **3**, Caussette and the cited art above teach a device substantially as claimed.

With respect to the recitations "which has a sensitivity of 2 ppm" (claim 2) or "which has a sensitivity or 6-10 ppm for detecting monensin in the liquid sample" (claim 3), applicant is reminded that a limitation of the claimed invention must result in a structural difference between the claimed invention and the prior art in order to patentably distinguish the claimed invention from the prior art. If the prior art structure is capable of achieving the limitation, then it meets the claim. Applicant is also reminded that claim scope is not limited by claim language that does not limit a claim to a particular structure. See MPEP 2111.04. In this case, no clear structural differences are invoked by the recitation of the sensitivity as claimed. Because there is not distinction between the claimed structure and that of the cited prior art, it is the case that the lateral flow device of Caussette would similarly be expected capable of achieving the same sensitivity. The limitations of claims 2 and 3 fail to clearly require any additional reagent of component be part of the device and fails to impart any particular structural feature or limitation.

Regarding **claims 4** and **6**, Caussette teaches that the sample receiving section is made of a porous material (Caussette, column 7, lines 29-30). Caussette further teaches that porous material refers to any material capable of providing lateral flow, for example polymeric materials such as polyester (claim 4), also cotton, glass fiber, nitrocellulose and blends of nitrocellulose with polymeric materials (claim 6; Caussette, column 9, lines 31-35).

Regarding **claim 7**, Caussette teaches labeling the antibiotic binding protein with colloidal gold particles (Caussette, column 5, lines 27-28).

Regarding **claim 8**, Caussette teaches covalently conjugating the capture reagent to the detection region through a carrier such as bovine serum albumin, optionally coupling the capture reagent to the carrier by a spacer (Caussette, column 8, lines 49-55).

Regarding **claim 9**, Caussette teaches labeling the control reagent and further teaches that the control agent is IgY labeled with gold particles (Caussette, column 11, lines 46-47).

7. **Claim 5 is rejected under 35 U.S.C. 103 as being unpatentable over Caussette et al., in view of Watanabe et al., Berlina, Godoy-Navajas, and Janeway, as applied to claim 1, and further in view of Salminen et al., 2020, Analytical Chemistry.**

Regarding **claim 5**, Caussette and the cited art above teach a competitive lateral flow device for the detection of monensin substantially as claimed.

Caussette does not teach that the conjugate pad is a chopped glass pad.

Salminen teaches an optimized highly sensitive lateral flow device (Salminen, 'Abstract', lines 6-8). Salminen further teaches testing 10 different conjugate pads and choosing the Ahlstrom 8950 glass fiber conjugate pad (Salminen, page 15768, 3rd paragraph, lines 14-18). Salminen further teaches that Ahlstrom 8950 is made of chopped glass (Salminen, see supplemental material, page S-3, supplemental Table S2) and has the best signal/background ratio of the materials tested (Salminen, see supplemental material, page S-7, figure S5).

It would have been *prima facie* obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to have modified the device of Caussette with the chopped glass conjugate pad of Salminen because of the teaching of Salminen that the chopped glass conjugate pad resulted in the best signal/background ratio out of 10 tested conjugate pad materials.

The ordinary artisan would have a reasonable expectation of success because Salminen teaches success using the chopped glass conjugate pad in a lateral flow device, the same device as taught by Caussette.

8. **Claim 22 is rejected under 35 U.S.C. 103 as being unpatentable over Caussette et al., in view of Watanabe et al., Berlina, Godoy-Navajas, and Janeway, as applied to claim 1, and further in view of Mount et al., 1978, Journal of the Association of Official Analytical Chemists.**

Regarding **claim 22**, Caussette and the cited art above teach a competitive lateral flow device for the detection of monensin substantially as claimed.

Caussette teaches that compounds facilitating resuspension and/or dissolution and/or mixing of the labeled antibiotic binding protein, the labeled control reagent and the sample can be added, suitable compounds include a buffer, such as Tris buffer (Caussette, column 3, lines 27-35).

Caussette and the cited art fail to teach an aqueous extraction buffer comprising phosphate buffered saline and 1% Tween-20.

Mount teaches an enzyme immunoassay for determination of monensin in biological samples (Mount, see title). Mount further teaches that the samples and monensin standards are diluted in PBS Tween (Mount, page 201, Figure 2, legend line 6). Mount further teaches that the phosphate-buffered saline is PBS containing 0.05% tween-20 (Mount, page 202, see '(g)').

It would have been *prima facie* obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to have modified the device of Caussette in order to use PBS 0.5% Tween-20 in place of the Tris buffer of Caussette as an obvious matter of a simple substitution of one art recognized buffer over another, both recognized as suitable for the same purpose, both suitable for diluting a sample for an immunoassay. The ordinarily skilled artisan would have been

motivated to do so, because the buffer of Mount performs the same function as specified in the claim in substantially the same way and produces substantially the same results of diluting a sample for detection of monensin by immunoassay. Substituting one buffer for applying the sample with another buffer, both of which are taught for detecting a sample by immunoassay would yield a predictable result.

The ordinarily skilled artisan would have had a reasonable expectation of success in doing so because of the teaching of Mount that demonstrates success of detecting monensin in a sample by immunoassay, as a result, one would have a reasonable expectation of success, substituting Caussette's Tris buffer with another buffer known to be used in immunoassay detection of monensin.

Regarding the limitation 1% Tween-20 as presently recited in the claims; it has been previously established by the courts that a prima facie case of obviousness exists where the claimed ranges or amounts do not overlap with the prior art but a merely close (see MPEP 2144.05). In the present case, 1% Tween-20 is an amount substantially close to 0.05% as taught by Mount and as such it would have been considered *prima facie* obvious in view of Mount to have arrived at the presently claimed Tween-20 concentration out of the course of routine experimentation, in order to achieve a desired effect of suspending a sample comprising monensin in a liquid for application on a lateral flow device.

The courts have also held that a particular parameter, when recognized as a result effective variable, i.e., a variable that achieves a recognized result, may be subject to routine optimization in order to determine optimum or workable ranges of said variable (also MPEP 2144.05). In the present case it would have been prima facie obvious to one of ordinary skill in the art to determine the optimum Tween-20 concentration by optimizing what was known in the art to be a suitable concentration of Tween-20 as taught by Mount. Mount teaches that it was routine in the art to use a Tween-20 concentration of 1%. It would have been obvious to have arrived at the claimed concentration of 1% Tween-20 as an obvious matter of routine experimentation because the Tween-20 concentration is an

art recognized result effective variable (namely the variable which results in successful detection of monensin).

One of ordinary skill in the art would have a reasonable expectation of success because a Tween-20 concentration of 0.05% is substantially similar to 1%. One of ordinary skill would expect success optimizing by routine experimentation because such optimization would allow the operator to achieve the desired detection of monensin and further because optimization is well within the skill level of the ordinary artisan.

Response to Arguments

9. Applicant's arguments filed 06/20/2024 have been fully considered but they are not persuasive. Regarding Applicant's arguments starting on page 4 that Lin and Lee provide neither a motivation nor a reasonable expectation of success in making an LFA device for monensin detection, the current rejection does not rely on Lin nor Lee and therefore the argument is moot.

For this reason, the arguments are not persuasive.

Communication

10. Any inquiry concerning this communication or earlier communications from the examiner should be directed to STEFANIE J KIRWIN whose telephone number is (571)272-6574. The examiner can normally be reached Monday - Friday 7.30 - 4 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Gregory Emch can be reached on (571) 272-8149. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/STEFANIE J. KIRWIN/
Examiner, Art Unit 1677

/Jeffrey T. Palenik/
Acting SPE, Art Unit 1677